
Sketch- and Text-Guided Image Generation with Diffusion Models

Quyen Tran
DePauw University
quyentran_2027@depauw.edu

Abstract

This work investigates diffusion-based image generation from sketch and text inputs. Using the DDPM framework, the denoising process is conditioned on a Canny-edge sketch, a CLIP text embedding, and the diffusion timestep. Three backbone architectures (U-Net, ResNet, and DiT) are trained under identical settings to isolate architectural differences. Due to computational constraints, training is performed at a small resolution and for a limited number of epochs. As a result, the outputs are preliminary but still informative. Qualitative results indicate that AE and VAE models tend to produce blurred images, while diffusion-based approaches preserve more structural detail from the sketch and capture richer color variations from the caption. These early findings suggest the potential of sketch- and text-guided diffusion models for producing images with more artist-like structure and interpretability. Code is available at https://github.com/quyenlhttran/final_project.

1 Introduction

How can an image be created in a way that feels more human? When people draw, they rarely start with colors or fine details. They usually begin with a sketch to define the structures and then fill in with the colors later. However, most image generation models try to paint and draw at the same time, and they often produce results that are visually detailed but feel artificial and mechanical [3, 28, 42]. This contrast leads to an important question: can we train a model to learn to sketch before painting so that its creations feel more human in form and intent?

Generative models have been used widely for creating images across a range of tasks, from unconditional synthesis to structured image-to-image translation. Early work based on variational autoencoders (VAEs) [20] introduced a probabilistic framework for modeling image distributions through continuous latent representations. GANs [9] provided an alternative approach based on adversarial training, and subsequent architectures such as DCGAN [22], Pix2Pix [16], CycleGAN [46], and StyleGAN [19] demonstrated that these models can produce images with detailed textures, natural color variation, and consistent structural forms.

More recently, diffusion models have emerged as a powerful alternative for image generation [7, 34]. DDPMs [15] formulate generation as a gradual denoising process, producing images by reversing a forward noising schedule. Conditional diffusion models, including image-to-image variants such as Palette [31], extend this framework to tasks that modify appearance while preserving spatial structure. As a result, diffusion models have become widely adopted in both research and practical systems [8, 32, 40].

Therefore, this work examines how diffusion models use sketch and text guidance, with a focus on how architectural choices influence this process. It aims to answer the following questions:

- How do diffusion models interpret sketches as structural guidance, and how well do they preserve the geometry present in the input?

- How do sketch and text conditioning interact during generation, particularly in shaping color, texture, and overall appearance?
- How do different generative architectures respond to the same sketch–caption inputs, and how do their outputs differ?

Taken together, these experiments aim to explore how different generative architectures combine geometric and semantic information, and what kinds of images they produce when guided by both a sketch and a caption.

2 Related Work

2.1 Generative Models for Images

Generative image modeling has been studied extensively using frameworks such as variational autoencoders (VAEs) [20], generative adversarial networks (GANs) [9], and more recently diffusion models [15, 34]. While VAEs and GANs enable efficient one-shot generation, diffusion models formulate synthesis as a gradual denoising process and have demonstrated strong performance in conditional image generation. This work focuses on conditional diffusion models and study how architectural choices in the denoising network affect sketch- and text-guided generation.

2.2 Conditional and Image-to-Image Generation

Conditional extensions of VAEs, GANs, and diffusion models incorporate auxiliary inputs such as labels, images, or spatial maps to guide generation. In the GAN setting, models such as Pix2Pix [16] and CycleGAN [46] adapt adversarial training to paired and unpaired image-to-image translation, and have been applied to tasks including colorization, style transfer, and domain adaptation.

Across applications, including photo editing [1, 18, 4], colorization [44], texture transfer [35], these methods usually handle structure and appearance at the same time in one forward pass. This often produces detailed images, but it does not clearly separate the step of defining the structure from the step of adding appearance.

Diffusion-based methods have been adapted in a similar spirit to handle conditional and image-to-image tasks. Variants of latent diffusion [29] and related models treat text, segmentation masks, or input images as conditioning signals to control the denoising process, supporting applications such as text-to-image synthesis, inpainting, super-resolution, and structure-preserving img2img transformations within a unified framework.

2.3 Sketch-Based and Sketch-Conditioned Generation

Sketch-based image synthesis uses sparse line drawings as structural constraints for controllable generation. Early work on sketch modeling includes sequence-based representations such as SketchRNN [10], while subsequent approaches investigate convolutional and transformer-based architectures for raster sketch-to-image translation [17, 33, 38, 41]. In the supervised translation setting, sketches or edges are commonly treated as input modalities for sketch-to-photo or edge-to-image tasks, often using conditional GANs or related architectures.

2.4 Sketch- and Text-Guided Diffusion

Joint sketch–text conditioning combines geometric structure from sketches with semantic specification from text prompts. Several lines of work adapt pretrained text-to-image diffusion models to this setting. Image-guided diffusion methods such as ILVR [6] and SDEdit [23] treat images as additional guidance signals, while sketch-guided text-to-image diffusion [37] incorporates sketch-derived constraints alongside textual conditioning.

ControlNet [43] and T2I-Adapter [25] further extend this idea by adding control branches or adapters that condition on sketches and other spatial inputs, providing a general mechanism for multi-modal control. Related developments in text-rich and layout-aware generation refine the handling of structured visual constraints. DesignDiffusion [39] adapts a high-capacity text-to-image diffusion

backbone for design image generation, focusing on visual text accuracy, layout, and aesthetic quality through prompt enhancement, attention supervision, and preference-based fine-tuning.

In contrast to methods that primarily introduce new control branches, adapters, or guidance modules on top of large pretrained backbones, this work keeps the conditioning mechanism simple and fixed and instead varies only the denoising backbone architecture (U-Net [30], ResNet [11], DiT [26]). This design isolates the effect of backbone choice on sketch- and text-guided generation under a unified training and evaluation setup.

3 Methodology

This work employs the denoising diffusion probabilistic model (DDPM) framework [14] for conditional generation of RGB images from input sketches and text descriptions. Given a sketch $s \in \mathbb{R}^{1 \times H \times W}$, a text prompt y , and a target image $x_0 \in \mathbb{R}^{3 \times H \times W}$, the objective is to learn a conditional reverse diffusion process that reconstructs an image consistent with the provided conditioning inputs. An overview of the full conditioning pipeline is shown in Figure 1.

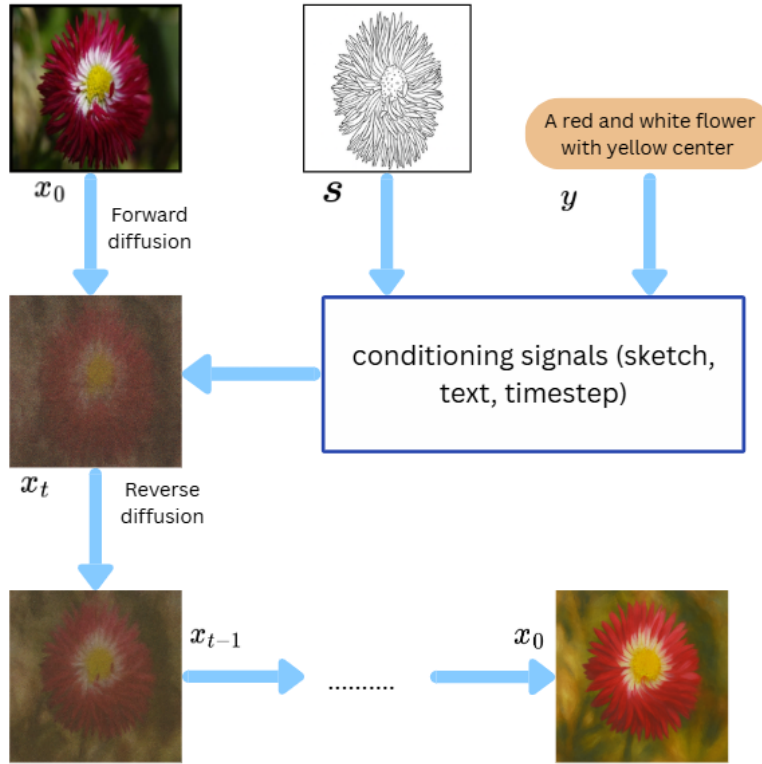


Figure 1: Overview of the sketch- and text-conditioned diffusion process. The target image x_0 is progressively noised to obtain x_t , while the reverse process reconstructs the image using conditioning signals from the sketch s , the text prompt y , and the timestep embedding.

3.1 Forward Diffusion Process

The forward diffusion process constructs a Markov chain that gradually corrupts a clean image x_0 by adding Gaussian noise over T timesteps. At each timestep t , a variance parameter $\beta_t \in (0, 1)$ controls the amount of noise injected. In practice, the sequence $\{\beta_t\}_{t=1}^T$ is chosen according to a fixed noise schedule. This project uses a linear schedule described in Section 4.1. Define $\alpha_t = 1 - \beta_t$, and let $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$ denote the fraction of the original signal preserved after t diffusion steps.

Given a noisy image x_{t-1} , the forward process produces x_t according to

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, \beta_t \mathbf{I}).$$

As t increases, repeated application of this transition progressively removes structure from the image. Because the composition of Gaussian transitions remains Gaussian, the distribution of x_t conditioned directly on the original image x_0 can be written in closed form as

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I}).$$

This expression makes it possible to generate x_t in a single step without simulating all intermediate states. As t increases, $\bar{\alpha}_t$ decreases and the sample x_t becomes dominated by Gaussian noise, eventually approaching an isotropic normal distribution. This progression establishes a feasible and analytically simple starting point for the reverse generative process.

3.2 Conditioning Mechanism

To enable controllable generation, the denoising model receives four conditioning signals: (i) the noisy image x_t , (ii) the sketch s , (iii) the timestep embedding e_t , and (iv) the text embedding e_y . Each signal provides a different form of information that guides the denoising process.

Noisy image (x_t). The noisy image x_t represents the current state of the diffusion process at timestep t . It contains a partially corrupted version of the target image and serves as the primary input that the model learns to denoise at each step.

Sketch (s). The sketch s provides explicit structural guidance and is extracted using Canny edge detection [5]. Canny edge detection is a classical method that extracts strong intensity gradients and produces a sparse, line-based representation in which object boundaries are retained and most texture and color information is removed. Figure 2 shows an image and its corresponding Canny edge sketch.



Figure 2: Example of Canny edge extraction. Left: original image. Right: corresponding sketch produced by Canny edge detection.

Timestep embedding (e_t). The timestep t is encoded using sinusoidal embeddings, originally introduced in Transformer models [36]. These embeddings represent the noise scale at each diffusion step and ensure that nearby timesteps have similar representations, enabling the model to generalize smoothly across diffusion stages.

Text embedding (e_y). Text prompts are embedded using CLIP [27], a vision–language model that maps text and images into a shared semantic space. The resulting embedding encodes high-level semantic attributes such as object category and color, providing semantic guidance that complements the geometric information from the sketch.

Conditioning injection. The sketch s is concatenated channel-wise with the noisy image x_t , while the embeddings (e_t, e_y) are injected through Feature-wise Linear Modulation (FiLM)-style affine modulation in convolutional architectures and through additive token modulation in transformer-based models (discussed in detail in 3.4). This conditioning interface is kept fixed across all backbone architectures to isolate the impact of architectural differences.

3.3 Reverse Denoising Objective

The goal of the reverse process is to undo the corruption introduced by the forward diffusion. Ideally, one would sample from the true posterior $q(x_{t-1} | x_t)$, but this distribution is not directly computable because it depends on the unknown data distribution of natural images. To make sampling possible, the model introduces a learned approximation $p_\theta(x_{t-1} | x_t, s, e_y, t)$, which is trained to mimic the true reverse conditional while incorporating sketch and text conditioning.

Following the DDPM formulation, the denoising network is parameterized to directly predict the noise term ϵ added at timestep t ; it does not model a distribution that is sampled during training. This choice utilizes the closed-form forward process:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}).$$

Therefore, predicting ϵ implicitly provides information about both the original image x_0 and the optimal reverse-step mean. This parameterization allows training to proceed using a standard mean-squared error loss, enabling gradients to be computed via backpropagation without stochastic sampling during optimization.

The resulting training objective is the simplified denoising loss

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{x_0, t, \epsilon} \left[\|\epsilon - \epsilon_\theta(x_t, s, e_y, t)\|_2^2 \right].$$

To construct x_t , a timestep t is sampled, noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is drawn, and x_t is formed using the closed-form forward distribution. Minimizing this loss trains the network to predict the exact noise that produced x_t , enabling the reverse process to invert each diffusion step while respecting both the geometric structure from the sketch s and semantic information from the text embedding e_y .

3.4 Backbone Architectures

U-Net is commonly used as a backbone in diffusion models, while other convolutional and transformer-based architectures have also been considered. This work aims to explore how different backbone architectures affect the denoising process under a unified diffusion framework. Thus, this work does not aim to perform an ablation of architectural components; instead, it compares representative backbone families under a fixed conditioning interface. A more systematic ablation of architectural design choices is left for future work.

Therefore, U-Net, ResNet, and DiT are selected for comparison, where ResNet represents a convolutional architecture and DiT represents a transformer-based architecture. All backbones share the same conditioning inputs but differ in how the denoising function is parameterized. The following paragraphs describe each architecture in detail.

U-Net Architecture. The U-Net is a multi-scale encoder-decoder with skip connections. The concatenated tensor $[x_t, s]$ is processed through downsampling and upsampling paths. The text and timestep embeddings (e_y, e_t) are first combined and projected through a learned linear layer to produce conditioning vectors, which are then used to modulate intermediate convolutional feature maps via channel-wise affine transformations. This modulation is applied at multiple resolutions, allowing conditioning information to influence both coarse global structure and fine local details throughout the network.

ResNet Architecture. The ResNet backbone consists of a single-resolution stream of residual blocks. Each block applies Feature-wise Linear Modulation (FiLM), a conditioning mechanism that modulates intermediate feature activations via channel-wise affine transformations. Formally, given a feature map h , FiLM applies $\text{FiLM}(h) = \gamma \odot h + \beta$, where the scale γ and shift β are not fixed model parameters. Instead, they are dynamically generated from the combined text-timestep embedding. Specifically, $(\gamma, \beta) = f_\theta(e_y, e_t)$, where f_θ is a learned linear projection or small multilayer perceptron trained jointly with the rest of the network. This modulation allows conditioning signals to influence the residual updates without altering the underlying convolutional structure. Unlike U-Net, this design does not incorporate multi-scale feature processing.

DiT Architecture. The DiT backbone tokenizes the concatenated input $[x_t, s]$ into non-overlapping patches and embeds them as a sequence of latent tokens. Positional encodings and a projected version of the text–timestep embedding are added to the tokens before they are processed by a transformer encoder. The transformed token sequence is then reshaped back into the spatial domain to predict the noise estimate $\epsilon_\theta(x_t, s, e_y, t)$. This provides a fundamentally different, attention-based instantiation of the denoising function.

3.5 Reverse Sampling

At inference time, the learned denoising model is used to iteratively sample an image from noise. Sampling begins from pure Gaussian noise $x_T \sim \mathcal{N}(0, \mathbf{I})$ and proceeds by iteratively evaluating learned reverse conditionals

$$p_\theta(x_{t-1} | x_t, s, e_y, t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, s, e_y, t), \sigma_t^2 \mathbf{I}), \quad t = T, \dots, 1.$$

To express the reverse mean in the ϵ -parameterization, the model predicts the noise $\epsilon_\theta(x_t, s, e_y, t)$. To specify the variance schedule, the formulation sets $\alpha_t = 1 - \beta_t$ and defines the cumulative product $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$. With these definitions, the reverse mean takes the form

$$\mu_\theta(x_t, s, e_y, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, s, e_y, t) \right).$$

To generate x_{t-1} , the sampling procedure chooses the update

$$x_{t-1} = \mu_\theta(x_t, s, e_y, t) + \sigma_t z, \quad z \sim \mathcal{N}(0, \mathbf{I}) \text{ for } t > 1.$$

At $t = 1$, the sampler uses $z = 0$ to produce a deterministic final image. The term σ_t is chosen as $\sqrt{\beta_t}$ to mirror the noise level used in the forward diffusion process.

Each reverse step subtracts the predicted noise, rescales the intermediate result according to the variance schedule, and then injects Gaussian noise to preserve sample diversity. Repeating this procedure from $t = T$ down to $t = 1$ gradually transforms the initial Gaussian sample into an RGB image aligned with the sketch structure and the caption semantics.

4 Experiments

This section describes the implementation details, training configuration, and evaluation protocol used to compare diffusion models with different backbone architectures alongside additional generative baselines. Training and validation loss curves for all models are provided in Appendix B.

4.1 Implementation Details

Dataset. All experiments are conducted using the Flowers BLIP Captions Dataset [21], which contains flower images paired with automatically generated textual descriptions. The captions are produced by BLIP, a vision–language model trained for image captioning, and serve as text conditioning inputs for the diffusion models. Each sample consists of an RGB image and an associated caption.

Sketch Preprocessing. Sketches are extracted using Canny edge detection as described in Section 3.2. Each RGB image is converted to grayscale, smoothed with a Gaussian filter, and processed by a Canny edge detector to produce a single-channel edge map $s \in \mathbb{R}^{1 \times 128 \times 128}$. Both the edge map and the corresponding RGB image are resized to 128×128 and normalized to the range $[-1, 1]$.

Text Encoding. Captions are encoded using the CLIP text encoder [27]. Tokenization follows the CLIP vocabulary, and the resulting text embeddings are kept fixed during training and inference.

Diffusion configuration. All diffusion models use $T = 1000$ denoising steps with a linear variance schedule, where $\beta_1 = 10^{-4}$ and $\beta_T = 0.02$. Timestep embeddings are implemented using sinusoidal positional encodings introduced in the Transformer architecture [36]. The input image resolution is fixed at 128×128 .

Training configuration. Models are trained for 100 epochs using the Adam optimizer with a learning rate of $1e-4$ and a batch size of 64. The simplified DDPM objective is used for all diffusion architectures. No data augmentation, EMA updates, or architecture-specific tuning is applied. Training is performed on a single NVIDIA A100 GPU.

Baseline models. For comparison, three non-diffusion generative models are trained under the same dataset split and conditioning format: an Autoencoder (AE) [2], a Variational Autoencoder (VAE) [20], and a Conditional GAN (cGAN) [24]. The AE and VAE use convolutional encoder-decoder architectures, while the cGAN follows a standard generator-discriminator design.

For all baseline models, the sketch s is concatenated channel-wise with the image input. Text embeddings are incorporated by concatenation with the latent representation in the AE and VAE models, and through conditional normalization layers in the cGAN generator.

The AE is trained using an ℓ_2 reconstruction loss, which minimizes the pixel-wise squared difference between the generated output and the target image. The VAE is optimized using the standard evidence lower bound (ELBO), which combines a reconstruction term with a Kullback-Leibler divergence that regularizes the latent distribution toward a fixed prior. The cGAN is trained using an adversarial objective, which minimizes a discriminator-based classification loss, augmented with an additional reconstruction term. All baselines receive the same sketch and text inputs as the diffusion models.

4.2 Comparison

All models are evaluated under a unified sketch-to-image generation protocol. Diffusion models based on U-Net, ResNet, and DiT backbones are trained using the identical diffusion configuration described previously, including the same noise schedule, timestep embeddings, text embeddings, and denoising loss. The only distinction among them is the parameterization of the denoising network, enabling architectural effects to be isolated from all other factors.

Non-diffusion baseline models (AE, VAE, and cGAN) are trained on the same training split and evaluated on the same validation set, using identical image resolution and normalization. For each sketch-caption pair in the validation set, every model produces a single RGB output image. Diffusion models generate samples by running the full reverse diffusion process from $T = 1000$ to $t = 1$ with identical sampling hyperparameters and a fixed random seed, while non-diffusion baselines generate outputs in a single forward pass.

To quantify performance, three complementary evaluation metrics are used: Fréchet Inception Distance (FID) for distribution-level realism [13], LPIPS for perceptual similarity [45], and CLIPScore for text-image semantic alignment [12]. Together, these metrics capture differences in visual quality, perceptual fidelity, and conditioning effectiveness across architectures. Detailed definitions and implementation details for all metrics are provided in Appendix A.

5 Results

The experimental findings are presented through both qualitative and quantitative analyses to evaluate how different generative architectures respond to sketch and text conditioning. The qualitative results examine visual fidelity and structural coherence on representative inputs, while the quantitative results assess perceptual similarity, distributional realism, and text-image alignment across all models.

5.1 Qualitative Results

A qualitative comparison is performed to assess how each architecture translates the same sketches and captions into images. Figure 3 shows representative outputs on a fixed subset of inputs for direct visual comparison of reconstruction quality, color usage, structural coherence, and alignment with the text.

Given the simple approach and constrained training budget, the outputs remain preliminary. The AE and VAE baselines produce smooth but blurry images with muted colors, often resembling degraded photographic reconstructions. The conditional GAN introduces greater chromatic variation but typically fails to recover coherent floral shapes.

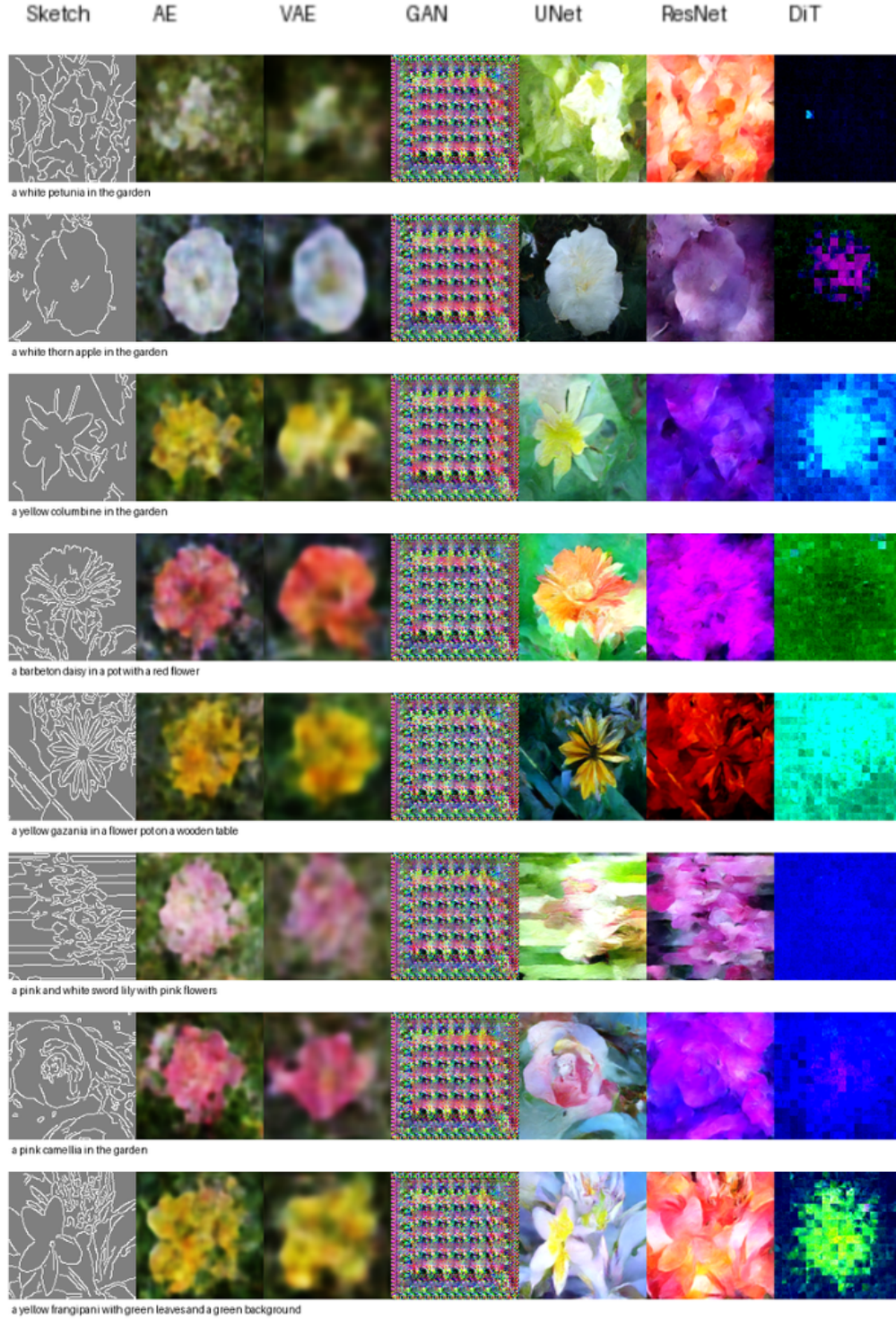


Figure 3: Qualitative comparison of all baselines and diffusion variants on representative sketch-caption pairs from the Flowers-BLIP dataset.

In contrast, the diffusion-based models demonstrate clearer structural cues from the sketches and generate colors that more closely follow the captions. Their outputs tend toward stylized, hand-drawn interpretations, indicating that diffusion processes better preserve global structure even under limited

training. Nonetheless, all diffusion variants still lack sharp details, suggesting that extended training or higher-resolution setups would further improve their qualitative performance.

5.2 Quantitative Results

Table 1: Quantitative comparison of all models. Lower is better for FID and LPIPS; higher is better for CLIP. **Bold** indicates the best value for each metric.

Model	FID ↓	LPIPS ↓	CLIP ↑
AE	388.59	0.8009	0.1957
VAE	483.78	0.8763	0.1954
GAN	315.59	0.9171	0.1915
Diffusion (U-Net)	362.78	0.8613	0.1958
Diffusion (ResNet)	417.00	0.9184	0.1922
Diffusion (DiT)	314.06	0.9237	0.1871

Table 1 reports FID, LPIPS, and CLIP scores for all models. Across diffusion architectures, performance varies substantially depending on the denoising backbone. Diffusion (DiT) achieves the lowest FID (314.06), indicating the greatest distribution-level realism according to this quantitative metric. Diffusion with U-Net and ResNet backbones obtains higher FID values (362.78 and 417.00, respectively), suggesting reduced effectiveness in matching the data distribution under the same training setup.

In terms of perceptual similarity, diffusion models exhibit higher LPIPS values (0.8613–0.9237) compared to some simpler baselines, indicating that while diffusion captures global structure well, it does not consistently preserve fine-grained perceptual details at the current resolution and training budget. Among diffusion variants, the U-Net backbone achieves the lowest LPIPS, while DiT and ResNet exhibit higher perceptual distances.

Semantic alignment measured by CLIP remains competitive across diffusion architectures, with diffusion (U-Net) achieving the highest CLIP score (0.1958). Overall, these results suggest that diffusion models are effective at integrating sketch structure and textual semantics. In particular, transformer-based diffusion (DiT) seems to be better in matching the overall image distribution, while convolutional backbones demonstrate relatively stronger semantic alignment under limited training.

6 Conclusion

In conclusion, this work explored how diffusion models integrate sketch and text conditioning for controllable image generation, with a focus on isolating the effects of different denoising backbones. Using a unified training and evaluation setup, diffusion models with three different backbone architectures (U-Net, ResNet, and DiT) were compared against AE, VAE, and GAN baselines on the Flowers BLIP Captions dataset.

Qualitative results show that diffusion models better preserve the structural cues present in sketches and produce color patterns that more closely follow the captions, even under limited training. In contrast, AE and VAE baselines tend to generate smooth but blurry images, while the GAN improves color richness but struggles to recover coherent shapes. Quantitatively, diffusion models achieve competitive FID scores but do not consistently improve LPIPS, suggesting strong global realism but limited perceptual fidelity under this training setup.

Overall, the findings highlight that sketch- and text-guided diffusion models can generate images with clearer structure and more interpretable conditioning behavior than non-diffusion baselines, but fully realizing their potential likely requires more extensive training, longer training schedules, or the incorporation of more advanced conditioning mechanisms beyond the current simple approach.

References

- [1] Rameen Abdal, Yipeng Qin, and Peter Wonka. Image2stylegan: How to embed images into the stylegan latent space? In *Proceedings of the IEEE/CVF international conference on computer*

- vision*, pages 4432–4441, 2019.
- [2] Dor Bank, Noam Koenigstein, and Raja Giryes. Autoencoders. *Machine learning for data science handbook: data mining and knowledge discovery handbook*, pages 353–374, 2023.
 - [3] David Bau, Jun-Yan Zhu, Jonas Wulff, William Peebles, Hendrik Strobelt, Bolei Zhou, and Antonio Torralba. Seeing what a gan cannot generate. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4502–4511, 2019.
 - [4] David Bau, Hendrik Strobelt, William Peebles, Jonas Wulff, Bolei Zhou, Jun-Yan Zhu, and Antonio Torralba. Semantic photo manipulation with a generative image prior. *arXiv preprint arXiv:2005.07727*, 2020.
 - [5] John Canny. A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-8(6):679–698, 1986. doi: 10.1109/TPAMI.1986.4767851.
 - [6] Jooyoung Choi, Sungwon Kim, Yonghyun Jeong, Youngjune Gwon, and Sungroh Yoon. Ilvr: Conditioning method for denoising diffusion probabilistic models. *arXiv preprint arXiv:2108.02938*, 2021.
 - [7] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021.
 - [8] Ben Fei, Zhaoyang Lyu, Liang Pan, Junzhe Zhang, Weidong Yang, Tianyue Luo, Bo Zhang, and Bo Dai. Generative diffusion prior for unified image restoration and enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9935–9946, 2023.
 - [9] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks, 2014. URL <https://arxiv.org/abs/1406.2661>.
 - [10] David Ha and Douglas Eck. A neural representation of sketch drawings, 2017. URL <https://arxiv.org/abs/1704.03477>.
 - [11] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.
 - [12] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. In *Proceedings of the 2021 conference on empirical methods in natural language processing*, pages 7514–7528, 2021.
 - [13] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
 - [14] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Proceedings of the 34th International Conference on Neural Information Processing Systems, NIPS ’20*, Red Hook, NY, USA, 2020. Curran Associates Inc. ISBN 9781713829546.
 - [15] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *arXiv preprint arxiv:2006.11239*, 2020.
 - [16] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A Efros. Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1125–1134, 2017.
 - [17] Moritz Kampelmühler and Axel Pinz. Synthesizing human-like sketches from natural images using a conditional convolutional decoder. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2020.

- [18] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. Analyzing and improving the image quality of stylegan. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8110–8119, 2020.
- [19] Tero Karras, Samuli Laine, and Timo Aila. A style-based generator architecture for generative adversarial networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(12): 4217–4228, 2021. doi: 10.1109/TPAMI.2020.2970919.
- [20] Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2014.
- [21] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International conference on machine learning*, pages 12888–12900. PMLR, 2022.
- [22] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *Proceedings of International Conference on Computer Vision (ICCV)*, December 2015.
- [23] Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, and Stefano Ermon. Sdedit: Guided image synthesis and editing with stochastic differential equations. *arXiv preprint arXiv:2108.01073*, 2021.
- [24] Mehdi Mirza and Simon Osindero. Conditional generative adversarial nets. *arXiv preprint arXiv:1411.1784*, 2014.
- [25] Chong Mou, Xintao Wang, Liangbin Xie, Yanze Wu, Jian Zhang, Zhongang Qi, and Ying Shan. T2i-adapter: learning adapters to dig out more controllable ability for text-to-image diffusion models. In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence, AAAI’24/IAAI’24/EAAI’24*. AAAI Press, 2024. ISBN 978-1-57735-887-9. doi: 10.1609/aaai.v38i5.28226. URL <https://doi.org/10.1609/aaai.v38i5.28226>.
- [26] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4195–4205, 2023.
- [27] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021.
- [28] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *International conference on machine learning*, pages 8821–8831. Pmlr, 2021.
- [29] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [30] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015.
- [31] Chitwan Saharia, William Chan, Huiwen Chang, Chris Lee, Jonathan Ho, Tim Salimans, David Fleet, and Mohammad Norouzi. Palette: Image-to-image diffusion models. In *ACM SIGGRAPH 2022 conference proceedings*, pages 1–10, 2022.
- [32] Chitwan Saharia, Jonathan Ho, William Chan, Tim Salimans, David J Fleet, and Mohammad Norouzi. Image super-resolution via iterative refinement. *IEEE transactions on pattern analysis and machine intelligence*, 45(4):4713–4726, 2022.
- [33] Leo Sampaio Ferraz Ribeiro, Tu Bui, John Collomosse, and Moacir Ponti. Sketchformer: Transformer-based representation for sketched structure. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14141–14150, 2020. doi: 10.1109/CVPR42600.2020.01416.

- [34] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *International conference on machine learning*, pages 2256–2265. pmlr, 2015.
- [35] Dmitry Ulyanov, Vadim Lebedev, Andrea Vedaldi, and Victor Lempitsky. Texture networks: Feed-forward synthesis of textures and stylized images. *arXiv preprint arXiv:1603.03417*, 2016.
- [36] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- [37] Andrey Voynov, Kfir Aberman, and Daniel Cohen-Or. Sketch-guided text-to-image diffusion models, 2022. URL <https://arxiv.org/abs/2211.13752>.
- [38] Fang Wang, Le Kang, and Yi Li. Sketch-based 3d shape retrieval using convolutional neural networks. pages 1875–1883, 06 2015. doi: 10.1109/CVPR.2015.7298797.
- [39] Zhendong Wang, Jianmin Bao, Shuyang Gu, Dong Chen, Wengang Zhou, and Houqiang Li. Designdiffusion: High-quality text-to-design image generation with diffusion models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 20906–20915, 2025.
- [40] Bin Xia, Yulun Zhang, Shiyin Wang, Yitong Wang, Xinglong Wu, Yapeng Tian, Wenming Yang, and Luc Van Gool. Diffir: Efficient diffusion model for image restoration. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 13095–13105, 2023.
- [41] Pengfei Xu, Banhuai Ruan, Youyi Zheng, and Hui Huang. Sketchformer++: A hierarchical transformer architecture for vector sketch representation. In *Computational Visual Media: CVM 2024*, page 24–41, Berlin, Heidelberg, 2024. Springer-Verlag. ISBN 978-981-97-2094-1. doi: 10.1007/978-981-97-2095-8_2. URL https://doi.org/10.1007/978-981-97-2095-8_2.
- [42] Shamim Yazdani, Akansha Singh, Nripsuta Saxena, Zichong Wang, Avash Palikhe, Deng Pan, Umapada Pal, Jie Yang, and Wenbin Zhang. Generative ai in depth: A survey of recent advances, model variants, and real-world applications. *Journal of Big Data*, 12(1):1–43, 2025.
- [43] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models, 2023. URL <https://arxiv.org/abs/2302.05543>.
- [44] Richard Zhang, Phillip Isola, and Alexei A Efros. Colorful image colorization. In *European conference on computer vision*, pages 649–666. Springer, 2016.
- [45] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595, 2018.
- [46] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. pages 2242–2251, 10 2017. doi: 10.1109/ICCV.2017.244.

Appendix

A Quantitative Evaluation Metrics Details

This section provides additional details on the quantitative metrics used to evaluate sketch- and text-guided image generation. Three standard metrics are employed: Fréchet Inception Distance (FID), Learned Perceptual Image Patch Similarity (LPIPS), and CLIPScore. All three metrics rely on pretrained neural networks, which introduces model-dependent bias; however, prior work has shown that while absolute values may vary across feature extractors or pretrained models, relative rankings tend to remain stable when methods are evaluated consistently on the same dataset. Consequently, these metrics are best interpreted comparatively rather than as absolute measures of quality.

Fréchet Inception Distance (FID). FID [13] measures the similarity between the distributions of generated images and real images in the feature space of a pretrained Inception network. In practice, both real and generated images are passed through a fixed Inception-V3 model, and feature vectors are extracted from an intermediate pooling layer. These feature representations are then modeled as multivariate Gaussian distributions.

Let (μ_r, Σ_r) denote the mean and covariance of features extracted from real images, and (μ_g, Σ_g) denote those extracted from generated images. The Fréchet Inception Distance is defined as

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right).$$

The term ‘‘Fréchet’’ refers to this closed-form distance between two Gaussian distributions, which accounts for differences in both their means and covariance structures. Lower FID values indicate closer agreement between the generated and real image distributions.

Learned Perceptual Image Patch Similarity (LPIPS). LPIPS [45] measures perceptual similarity between two images based on distances in a deep feature space rather than pixel-level error. Given a ground-truth image x and a generated image $\hat{x}$, both images are passed through a pretrained convolutional neural network such as VGG, which is trained on ImageNet and commonly used as a generic feature extractor. The difference between their normalized feature activations is then aggregated as

$$\text{LPIPS}(x, \hat{x}) = \sum_l w_l \|\phi_l(x) - \phi_l(\hat{x})\|_2^2,$$

where ϕ_l denotes the feature map at layer l and w_l are learned weights. Lower LPIPS values correspond to closer perceptual similarity and better structural fidelity between images.

CLIPScore. CLIPScore [12] evaluates text–image semantic alignment using CLIP’s image and text encoders [27]. Given a caption y and a generated image $\hat{x}$, the score is defined as the cosine similarity between their embeddings:

$$\text{CLIPScore}(y, \hat{x}) = \frac{\langle f_{\text{text}}(y), f_{\text{image}}(\hat{x}) \rangle}{\|f_{\text{text}}(y)\| \|f_{\text{image}}(\hat{x})\|}.$$

Higher values indicate that the generated image better reflects the semantic content of the caption.

B Training Curves

Figure 4 summarizes the optimization behavior of all models during the 100-epoch training schedule used in our experiments. These curves provide additional insight into convergence speed, stability, and generalization across autoencoders, GANs, and diffusion backbones.

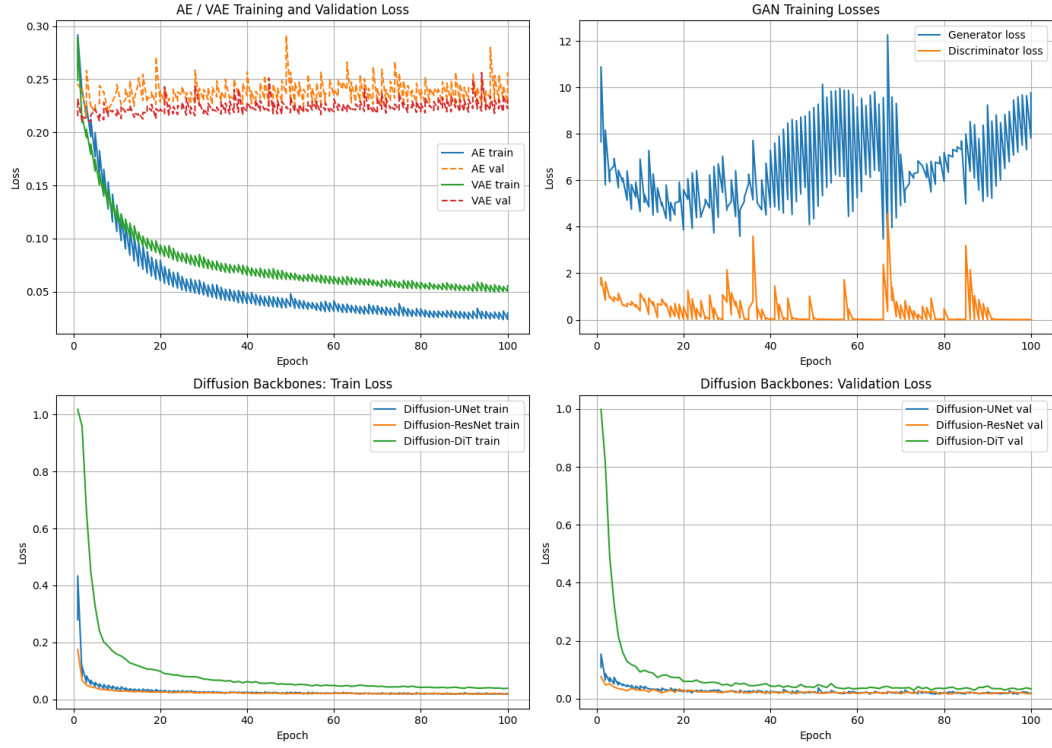


Figure 4: Training and validation loss curves for AE, VAE, GAN, and diffusion models (U-Net, ResNet, DiT).